Prevention of Pediatric Burn Injuries

Most of the pediatric burns are accidental and hence preventable. Most often it is the parent's carelessness or oversight which leads to accidental burns in children. The aim of burn prevention is a continuing reduction in the number of serious burn injuries, especially at home **(Table 1)**.

TABLE 1: Burn prevention.

Prevent fires at home	***Prevent burn injury at home***
☑ Install and use smoke detectors	☑ Do not leave hot water buckets unattended near children
☑ Keep fire, matches, and lighters out of the reach of children	☑ Mix hot water into cold water in a bucket while preparing to bathe the child
☑ Avoid cigarette smoking in bed	☑ Do not let children near hot irons
☑ Do not leave lit candles or diyas unattended	☑ Avoid using tablecloths which children can pull to avoid spillage of hot liquids on them
☑ Use flame-retardant-treated clothing	☑ Practice escape procedures with children for use in case of disasters
☑ Do not allow children in kitchen while cooking food	☑ Crawl beneath smoke if a fire occurs indoors
☑ Avoid floor-level cooking	
☑ Keep cloth items off heaters	
☑ Propagate "say no to crackers"	
☑ Do not allow children to play with firecrackers unattended at all	

Classification of Burn Injuries (Fig. 1)

Proper triage and treatment of burn injury require assessment of the extent and depth of the injury. The injuries are categorized according to the depth of skin affected.

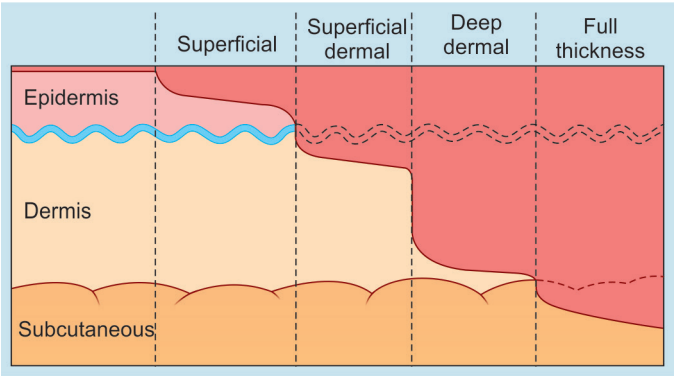


Fig. 1: Different burn depths.

First-degree Burns	
Caused by	☑ Flash burns and sunburns
Depth affected	☑ Only the epidermis is affected
Symptoms and signs	☑ Pain, burning, and erythema which blanches on pressure ☑ No blistering as tissue damage is usually very minimal
Healing	☑ Spontaneous, within 2–3 days, accompanied by peeling of burnt epithelium ☑ Leaves no scars

Second-degree Burns: Superficial Dermal (Figs. 2A to C)	
Caused by	☑ Contact burns, scalds, chemicals, electrical, and friction
Depth affected	☑ Entire epidermis and part of superficial dermis
Symptoms and signs	☑ Pink, wet, and glistening area which blanches on pressure ☑ Pain, swelling, erythema, and formation of blisters ☑ Extremely painful because a large number of remaining viable nerve endings are exposed
Healing	☑ Heal in 7–14 days as the epithelium regenerates in the absence of infection

Fig. 2A: Perineal burn wound caused by hot water.

Fig. 2B: Second-degree mixed superficial and deep dermal.

Fig. 2C: Second-degree superficial dermal burn.

Second-degree Burns: Deep Dermal (Figs. 2B and D)	
Caused by	☑ Contact burns, scalds, chemicals, and electrical
Depth affected	☑ Entire epidermis and deep dermis ☑ Subcutaneous tissue is intact
Symptoms and signs	☑ Pain is less (as superficial nerve endings are burnt) ☑ Dry, waxy white look, and no blisters
Healing	☑ Heal spontaneously (over 3–5 weeks) if wounds are kept clean and infection free ☑ Hypertrophic scarring is significant

Fig. 2D: Second-degree deep dermal burn.

Classification of Burn Injuries

Third-degree Burns or Full-thickness Burns (Fig. 2E)	
Caused by	☑ Electrical, chemical, and contact
Depth affected	☑ Entire skin and appendages
Symptoms and signs	☑ Variable presentation ☑ Marble white to brown to charred black ☑ Leathery and dry ☑ Little edema and no pain
Healing	☑ The wound cannot epithelialize and can heal only by wound contraction or skin grafting

Fig. 2E: Third-degree burn.

Estimation of Body Surface Area for a Burn

- ☑ The volume of fluid needed in resuscitation is calculated from the estimation of the extent and depth of the burn surface. Mortality and morbidity also depend on the extent and depth of the burn.
- ☑ The burn wound size is expressed as the percentage of body surface area (BSA) that is burnt.
- ☑ Only second- and third-degree burns are used to calculate total burn area.
- ☑ In small burns, <10% of BSA, *"rule of palm"* may be used, especially in outpatient settings (the area of patient's palm from the wrist crease to the fingertip with fingers closed in the child equals 1% of the child's BSA).

- ☑ *"Rule of nines"* may be used only in children >12 years old. The body is divided into anatomic areas, which constitute a surface area of 9% or its multiples **(Fig. 3)**.
- ☑ Lynch and Blocker's *"rule of fives"* works well for infants where head and neck, anterior and posterior trunk are 20% each (multiples of 5) and each limb is 10% **(Fig. 4)**.

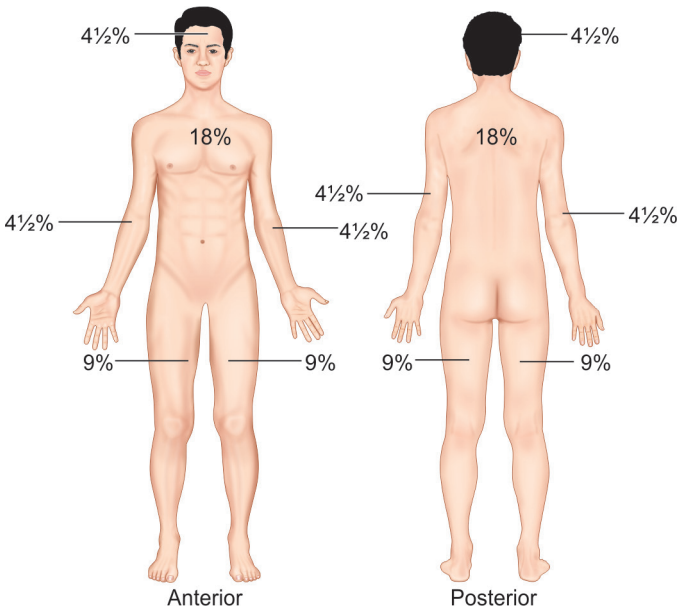


Fig. 3: Estimation of burn area by Wallace's "rule of nines".

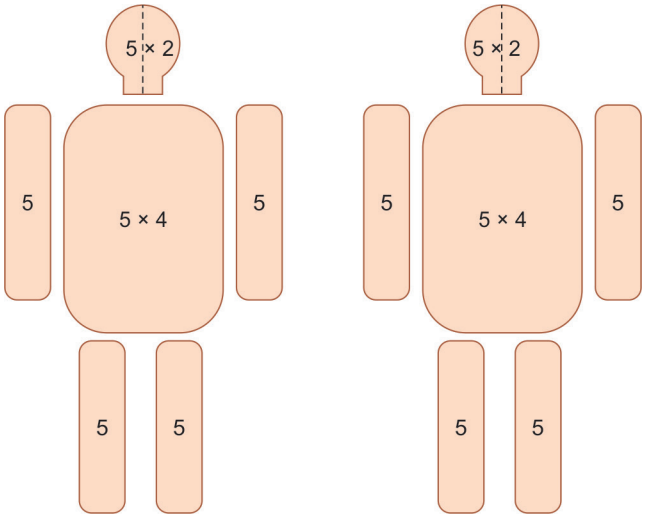


Fig. 4: Estimation of burn area by "rule of five" (Lynch and Blocker).

☑ *Lund and Browder chart*: Appropriate burn chart for different childhood age groups should be used to accurately estimate the extent of BSA burned **(Fig. 5)**.

Burn estimate and diagram age and area

Initial evaluation*

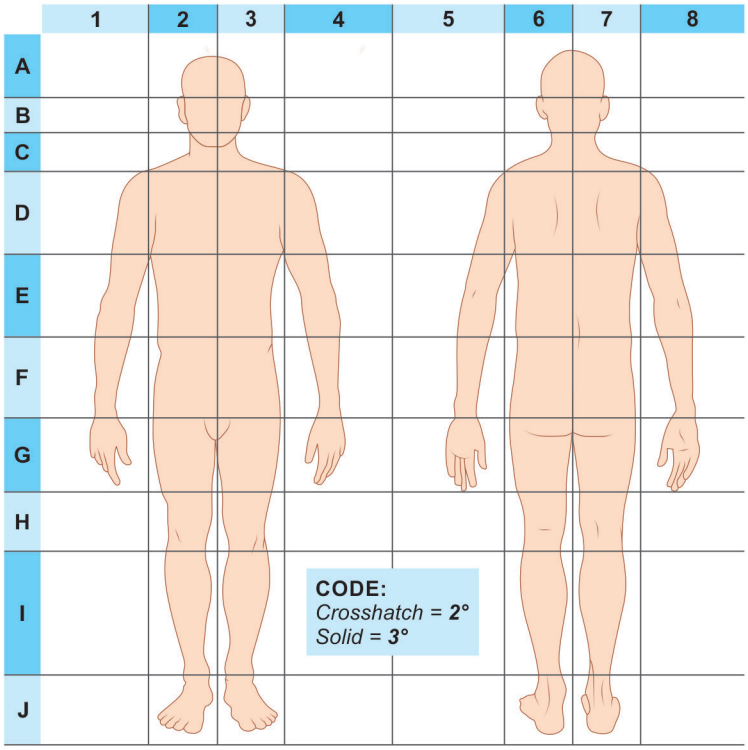
Signature: _____

Date of burn: _____

Date completed: _____

*To be completed by the admitting physician or licensed independent practitioner on admission

This is a working burn estimate diagram only, and is not as accurate as photography



Area	Birth-1 yr	1–4 yrs	5–9 yrs	10–14 yrs	15 yrs	Adult	2°	3°	Total
Head	9	17	13	11	9	7			
Neck	2	2	2	2	2	2			
Anterior trunk	13	13	13	13	13	13			
Posterior trunk	13	13	13	13	13	13			
Right buttock	2.5	2.5	2.5	2.5	2.5	2.5			
Left buttock	2.5	2.5	2.5	2.5	2.5	2.5			
Genitalia	1	1	1	1	1	1			
Right upper arm	4	4	4	4	4	4			
Left upper arm	4	4	4	4	4	4			
Right lower arm	3	3	3	3	3	3			
Left lower arm	3	3	3	3	3	3			
Right hand	2.5	2.5	2.5	2.5	2.5	2.5			
Left hand	2.5	2.5	2.5	2.5	2.5	2.5			
Right thigh	5.5	6.5	8	8.5	9	9.5			
Left thigh	5.5	6.5	8	8.5	9	9.5			
Right lower leg	5	5	5.5	6	6.5	7			
Left lower leg	5	5	5.5	6	6.5	7			
Right foot	3.5	3.5	3.5	3.5	3.5	3.5			
Left foot	3.5	3.5	3.5	3.5	3.5	3.5			
***Only 2° and 3° burns are included in the total TBSA burn percent						*Total*			

Fig. 5: Lund and Browder burn assessment chart (all numbers are in percentages). (TBSA: total body surface area)

Indications for Hospitalization

- ☑ Burns affecting >10% of BSA
- ☑ Third-degree burns >5% total body surface area (TBSA)
- ☑ Electrical burns caused by high-tension wires or lightning
- ☑ Chemical burns
- ☑ Inhalation injury, regardless of the amount of BSA burned
- ☑ Inadequate home or social environment
- ☑ Suspected child abuse or neglect
- ☑ Burns to the face, hands, feet, perineum, and genitals
- ☑ Burns in patients with preexisting medical conditions that may complicate the acute recovery phase
- ☑ Concomitant injuries (such as fractures and head injury)

Treatment

First Aid in Scalds and Burns

Effective first aid and triage can decrease both the extent (area) and the severity (depth) of injuries.

- ☑ Burnt area should be cooled immediately with tap water (preferably running water) for at least 10–15 minutes.
- ☑ In case of chemical burns, wash off the chemical by running water for at least 20 minutes or till litmus test is negative.
- ☑ Very cold water and ice should not be used.
- ☑ Do not immerse severe burns in water as it may lead to hypothermia.
- ☑ Oral drinks to be offered if there is delay.
- ☑ Do not try to puncture the blisters.
- ☑ Rings and bracelets to be removed from the affected area as they may cause constriction when edema occurs.
- ☑ Remove clothing from the burned area but do not pull the cloth if stuck.
- ☑ Cover burned area with a clean cloth before going to the hospital to prevent contamination and hypothermia.
- ☑ Do not apply toothpaste, gentian violet, or any other home remedy on the burned area as it makes evaluation of depth difficult.
- ☑ In case of electrical burn, switch off the mains supply before pulling child away from source of current.
- ☑ Check for signs of circulation, breathing, and movement, if none initiate cardiopulmonary resuscitation (CPR).

Treatment

Initial Assessment

- ☑ Assess general condition [airway, breathing, and circulation (ABC)], nature of burns, age, extent, depth, concomitant diseases, and associated injuries.
- ☑ Turn the patient to one side.
- ☑ Airway maintenance and suction, if any secretions.
- ☑ Assess for inhalation injury—history of fire in closed space, facial burns, singed nasal hairs, eyebrows, hoarseness of voice, facial edema, swelling of lips, and oral mucosa.
- ☑ Weigh the child.
- ☑ Insert an intravenous (IV) line in peripheral veins, if possible, but if in shock put in a central line.
- ☑ Allow oral fluids if patient is conscious and <10% burns.
- ☑ If >10% burns, nil orally to prevent abdominal distension.
- ☑ Foley's catheter to monitor urine output in all children with moderate-to-severe burns.

Fluid Resuscitation

- ☑ It is of utmost importance in first 24 hours. Compared to adults, children require IV fluid resuscitation for burns as small as 10–20% TBSA or smaller percentages with inhalation injuries and high-tension electrical injuries to ensure diuresis.
- ☑ Fluid of choice is Ringer's lactate for resuscitation along with daily maintenance in the form of dextrose saline.
- ☑ There are many formulae for calculation of fluid requirement. No single formula can be adhered to strictly in managing a burned child. The most important fact is not to overhydrate or under-resuscitate a child.
- ☑ A formula with ceiling of 50% TBSA is essential along with daily requirement; for this purpose modified Brook's formula satisfies all requirements.
- ☑ According to modified Brook's formula fluid requirement is calculated as shown in **Table 2**.

TABLE 2: Modified Brook's formula fluid requirement.

Time from burn	*Fluid requirement*
First 24 hours from burn	☑ 2 mL/kg weight/% BSA + daily maintenance ☑ Half of this in first 8 hours ☑ Second half in next 16 hours
Second 24 hours from burn	1 mL/kg weight/% BSA + daily maintenance

Treatment

Fluid Resuscitation

- ☑ Daily maintenance fluid is given in the form of N/2 dextrose saline or N/4 dextrose saline and is calculated as per the weight of the child **(Table 3)**.

TABLE 3: Daily maintenance fluid.

Weight of child	***Daily maintenance therapy in children***
0–10 kg	100 mL/kg
10–20 kg	1,000 mL + 50 mL/kg for every kg above 10 kg
>10 kg	1,500 mL + 20 mL/kg for every kg above 20 kg

- ☑ In first 24 hours, colloids are not indicated except if fluid requirement is very high in extensive burns to increase the oncotic pressure. Colloids are given in the form of 5% albumin or plasma in the volume of 0.3–0.5 mL/kg/% BSA.
- ☑ After 24 hours, colloids may be added only if plasma oncotic pressure is very low despite adequate fluid replacement by crystalloids or if serum protein falls below 4 g/dL.
- ☑ Adequacy of the fluid resuscitation is assessed by measuring urine output which should be at least 1 mL/kg/h and it is the most practical method in any kind of clinical setting.
- ☑ The other parameters to be monitored are the vitals, acid-base balance, mental status, and laboratory parameters.

Laboratory parameters: Hematocrit, serum electrolytes, urine osmolality, arterial blood gas (ABG).

Airway Management in Inhalation Injury

- ☑ If there are signs of inhalation injury thorough assessment of respiratory system needed and start humidified O_2.
- ☑ Advise chest X-ray and ABG.
- ☑ Carboxyhemoglobin estimation if carbon monoxide poisoning suspected. Treat with oxygen, early ambulation, suction, inhalation steroids, chest physiotherapy, and bronchodilators.
- ☑ Indications for endotracheal intubation include increasing stridor, hypoxia, inability to clear secretions, inadequate ventilation, and increased intracranial pressure from hypoxia and if partial pressure of arterial oxygen (PaO_2) is <80 mm Hg.
- ☑ Tracheostomy to be avoided because of complications. It is indicated in very severe oral burns which prevents endotracheal intubation.
- ☑ If carbon monoxide poisoning suspected, use 100% O_2 (hyperbaric oxygen).
- ☑ If hydrogen cyanide toxicity, IV sodium thiosulfate 125–250 mg/kg and hydroxo-cobalamin are useful.

Treatment

Adjunctive Management

- ☑ H2 receptor blockers such as ranitidine and antacids for prophylaxis against gastroduodenal erosions and ulcerations.
- ☑ *Tetanus prophylaxis*: If >10% burns and immunized give dT. If status not known or contaminated, give human tetanus immunoglobulins.
- ☑ *Sedation and pain control*:

It should be preceded by correction of hypovolemia and hypoxia as they are the causes of restlessness in burn patients.

Morphine sulfate 0.1–0.2 mg/kg IV is given 4 hourly. Once shock phase is over can switch over to oral analgesics like paracetamol. Pethidine, pentazocine, chloral hydrate, promethazine, and benzodiazepines are also useful. For change of dressings, IV or inhalation agents such as ketamine 0.2–0.3 mg/kg and nitrous oxide and air mixture (Entonox) can be used. For anxiety, lorazepam 0.05–0.1 mg/kg/dose every 6–8 hourly can be given.

Nutritional Support

- ☑ Calories and nitrogen requirements are increased in burns because of hypermetabolism along with normal growth requirements. Malnutrition causes impaired wound healing, reduced resistance to infection, and leads to cellular dysfunction.
- ☑ Enteral feeding in the form of clear liquids as early as 3–6 hours to be encouraged because it preserves the gastrointestinal integrity and reduces the incidence of bacterial transmigration across the gut. It should be increased gradually to all forms of liquids when patient is fully resuscitated.
- ☑ Tube feeding is indicated if child is unable to take orally. The feed should have 1 kcal/mL and low osmolarity (300–700 mOsmol/L). Start with one-fourth desired volume and increase at 5 mL/hour.
- ☑ Daily calorie and protein requirement can be estimated with following formula: 65 Kcal/kg + 35 kcal/% BSA for calories and 3 g/kg + 1 g/% BSA for proteins + sufficient vitamins/trace elements.
- ☑ Parental nutrition should be used only as a last resort if patient has vomiting, diarrhea, etc., because of complications such as metabolic abnormalities, sepsis, and immunosuppression.
- ☑ Multivitamin therapy with A, B, and C and trace elements such as zinc and magnesium to be added.

Treatment

Systemic Chemotherapy

- ☑ Burn wound is sterile, so if dressed with proper aseptic precautions no antibiotic cover is required early unless wounds are grossly contaminated. Wound and blood culture to be done periodically.
- ☑ Types of bacterial flora colonizing burn injuries are gram-positive (first week), gram- negative (second week), and fungal and antibiotic resistant organisms (third week) which need to be taken care of by appropriate antibiotics according to wound culture and sensitivity or unit antibiogram.

Wound Care

Burn wounds in children can be best treated by closed dressings except burns over the face which can be left open and perineum which is only covered with topical antimicrobial agent over which a diaper may be applied.

- ☑ Ambient room temperature must be kept between 28 and 30°C.
- ☑ Large blisters should be surgically debrided.
- ☑ Full thickness circumferential burns require escharotomy to improve circulation to distal extremities.
- ☑ Local antimicrobial agents to be used in dressing till wound epithelializes. They are:
  - 1% *silver sulfadiazine*: Once daily application enough, good penetration, both gram-positive and gram-negative coverage, does not stain clothes. Most easily available everywhere in India. Side effects—leukopenia which resolves spontaneously, not used below 2 months because it causes kernicterus, acid-base imbalance, and poor penetration into eschar.
  - *Silver nitrate 0.5%*: Bacteriostatic, used in second- and third-degree burns superficial penetration only, difficulty in dressing, causes staining and electrolyte imbalance.
  - *Mafenide acetate (11.1%)*: Deep and rapid penetration, useful for cartilage burns, best *Pseudomonas* coverage but is not available in India.
  - Other commonly used agents for small burns are bacitracin (good for face), Soframycin, and Neosporin. Other dressing agents are silver impregnated dressings, though expensive but are very good for deeper burns such as nanocrystalline silver dressings (ACTICOAT), Aquacel, and silver impregnated foams (Mepilex Ag). Their main advantage is the reduced frequency of dressing change because of sustained and prolonged release of silver.

Treatment

Biological Dressing

- ☑ These are temporary skin covers derived from various tissues. These are used if there is not enough native skin to cover and a likelihood that wound will epithelialize spontaneously, e.g., allografts, porcine grafts, and amnion and collagen sheets.
- ☑ Collagen dressings are very good for superficial burns as they decrease pain and evaporative losses, seal wound from environment, and encourage early healing.

Surgical Management in Burns

- ☑ Excisional surgery followed by skin grafting is the mainstay treatment for deep dermal and full thickness burns as it reduces morbidity, allows early recovery, reduces length of hospital stay, and reduces chances of burn sepsis and psychological trauma at such a young age.
- ☑ Early surgical excision is done on third to fifth postburn day.
- ☑ In case of extensive deep burns in which child cannot be taken up for excision, patient is treated conservatively by dressings and waiting for eschar to separate. Split thickness graft is applied once the area granulates.

Electric Injuries

- ☑ Electric burns might involve a small area but there is greater damage to muscles, nerves, and blood vessels.
- ☑ These children require more fluids because of deep tissue damage and massive edema.
- ☑ Treatment includes ABCs, immobilize spine, electrocardiogram (ECG), and renal function test (RFT).
- ☑ For minor burns dressing with topical antibiotics only is needed.
- ☑ For high tension burns (>1,000 volts), always admit. Diuresis should be achieved with the Ringer lactate and mannitol and alkalinize urine with IV sodium bicarbonate.
- ☑ High tension injuries require fasciotomy (to prevent muscle compartment syndrome) later followed by debridement and reconstruction procedures. Amputation of limbs may be required in case of onset of gangrene.

Treatment

Chemical Burns

- These can be due to alkali or acid. Dry eschar forms following acid burns and liquefaction necrosis following alkali burns.
- Alkali burns are usually deeper than acid burns. Early excision and skin grafting is needed in deep burns.

Further Reading

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Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Hyperthyroidism

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